

Advancing Mild Cognitive Impairment Detection: Integrating VR, MRI, and Neuropsychological Insights for Comprehensive Diagnosis

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Abstract

Identifying mild cognitive impairment (MCI) early is key to averting its progression into dementia, as MCI marks the intermediate phase between normal cognitive aging and dementia. Traditionally, MCI diagnosis has leaned on biomarkers such as magnetic resonance imaging (MRI) and neuropsychological evaluations. Yet, these methods come with their own set of challenges, sparking a growing interest in the use of virtual reality (VR) as a novel diagnostic tool. VR's ability to harness behavioral data from daily interactions has shown promise in accurately identifying MCI, although its practical application in clinical settings remains a topic of debate. This study delves into the relationship between VR and traditional biomarkers and assesses the advantage of incorporating VR into the diagnostic process for MCI. We engaged 54 participants who underwent neuropsychological tests, participated in VR test, and received MRI scans. Analysis through machine learning revealed that neuropsychological measures alone provided an accuracy of 90.7%. Yet, when VR biomarkers were combined with these measures, accuracy surged to 94.4%. This investigation highlights VR's potential as an effective biomarker in the early detection of MCI and suggests that its integration with conventional biomarkers can not only boost diagnostic precision but also reduce the financial and time burdens associated with traditional methods.

Keywords: Magnetic resonance imaging, Virtual reality, Neuropsychological test, Mild cognitive impairment, Early screening.

1. Introduction

Mild cognitive impairment (MCI) serves as a critical intermediary phase that bridges the gap between the natural cognitive decline associated with aging and the onset of dementia [1]. The timely identification of MCI is paramount in the endeavor to thwart the advancement of dementia, an ailment characterized by irreversible neurodegeneration [2]. Consequently, the pursuit of early MCI detection methodologies has intensified [3].

Traditionally, the diagnosis of MCI has relied heavily on the use of biomarkers, such as magnetic resonance imaging (MRI) and various neuropsychological evaluations [4, 5]. Despite their widespread adoption, these traditional diagnostic tools are not without their drawbacks, notably their significant financial and time commitments [6, 7], prompting a search for more accessible alternatives.

Emerging as a viable solution, the application of virtual reality (VR) technology stands out for its ability to gather detailed behavioral data from everyday activities [8, 9, 10]. By employing machine learning techniques to analyze this data, it becomes feasible to identify MCI in its nascent stages with remarkable precision. This approach not only boasts a high degree of accuracy but also benefits from being both time-efficient and cost-effective, typically requiring only about 15 minutes for assessment [11, 12].

Despite these advantages, the practicality of integrating VR-based diagnostics into routine clinical practice remains a topic of debate. This study aims to delve into this issue by juxtaposing traditional MCI biomarkers—MRI and neuropsychological testing—against VR-based methods. Our objective is to scrutinize the correlation between these diagnostic

tools and to evaluate the clinical relevance and potential advantages of incorporating VR technology into the MCI detection arsenal, both as a standalone tool and in conjunction with established biomarkers.

2. Materials and Methods

2.1. Participants

We recruited a total of 54 participants from Hanyang University Hospital, Republic of Korea, spanning from January 2022 to July 2023. The cohort of MCI patients was identified by two neurologists with 18 and 22 years of experience, who adhered to the diagnostic guidelines established by Albert and colleagues [13]. The inclusion criterion required the ability to process auditory and visual cues for interactive engagement within a VR setting. We excluded those with a history of dementia or who had undergone neurological surgeries. All participants were thoroughly briefed about the research procedures before the commencement of the study, and written consent was acquired from each. Ethical endorsement for this study was granted by the Institutional Review Board of Hanyang University Hospital, in alignment with the ethical principles of the Declaration of Helsinki (approval number HYUH-2021-08-020-004).

2.2. Neuropsychological tests

This study employed the Seoul Neuropsychological Screening Battery-C (SNSB-C) as the neuropsychological assessment tool, which is recognized for its validity within the Korean demographic [14]. The SNSB-C facilitated the evaluation of cognitive capabilities across five distinct domains: (1) Attention, gauged by the Digit Span Test: Backward (DST: B); (2) Language abilities, measured through the Short form of the Korean-Boston Naming Test (S-K-BNT); (3) Visuospatial skills, assessed via the Rey Complex Figure Test: Copyscore (RCFT: CS); (4) Memory function, evaluated with the Seoul Verbal Learning Test-Elderly's version: Delayed Recall (SVLT-E: DR); and (5) Frontal/executive function, determined by the Digital Symbol Coding (DSC) test. A professional psychological evaluator with 15 years of expertise, conducted all SNSB-C evaluations in this study.

2.3. Virtual reality biomarkers

In this study, we employed the virtual kiosk test, a VR-based assessment tool previously developed by our team, to gather VR biomarkers for early detection of MCI [11, 12]. This innovative test simulates the activity of placing an order at a virtual

kiosk, thereby enabling the collection of data indicative of early MCI stages. The hardware setup included a high-performance laptop powered by an Intel i7-12700H processor, 16 GB of RAM, and an NVIDIA GeForce RTX 3080 graphics card. For an immersive VR experience, participants were equipped with a head-mounted display featuring eye-tracking capabilities (HTC VIVE Pro Eye) and interacted with the virtual kiosk through a hand controller, while seated and monitored by two base stations (see Figure 1).

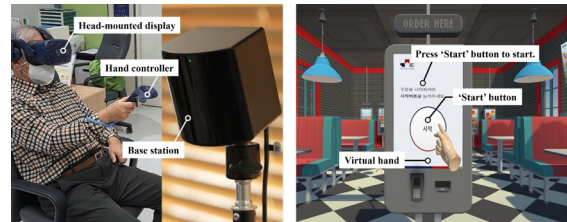


Figure 1: Setup of the virtual kiosk test (left) and the virtual kiosk and virtual hand (right)

The virtual kiosk task, depicted in Figure 2, involved a sequence of six tasks, bookended by ‘Start’ and ‘End’ phases: (1) selecting a dining venue; (2) choosing a burger; (3) selecting a side dish; (4) choosing a beverage; (5) selecting a payment method; and (6) inputting a four-digit payment code. Before beginning, participants received verbal instructions to order a specific meal (a shrimp burger, cheese sticks, and Coca-Cola) using a credit card, with ‘6289’ as the payment code. Throughout the test, we collected behavioral data to extract three key VR biomarkers. The first, ‘hand movement speed,’ was calculated by dividing the total hand movement distance by the test’s total duration. The second biomarker measured the proportion of time spent focusing on the specified menu items compared to all items, termed the ‘proportion of fixation duration.’ The third metric recorded was ‘the number of errors’ participants made during the task.



Figure 2: Six steps of the virtual kiosk test

2.4. Magnetic resonance imaging biomarkers

MRI imaging was performed at Hanyang University Hospital using a PHILIPS Ingenia CX 3T machine. The imaging technique employed was the 3D T1-weighted magnetization prepared rapid gradient echo with specific parameters: echo time (TE) and repetition time (TR) set at 2.9ms and 6.3ms respectively, a 9° flip angle, and a field of view of 256×256mm. The scans consisted of 211 slices, each with a voxel resolution of 1×1×1mm³.

The focus of our MRI analysis was on regions critical to memory function, namely the amygdala, hippocampus, and entorhinal cortex, due to their known involvement in MCI processes [15, 16]. To adjust for individual differences in overall brain size, we included the intracranial volume (ICV) in our analysis. The ICV measurement comprises the combined volume of the brain's white matter, gray matter, and cerebrospinal fluid. By normalizing MRI biomarkers against the ICV, we aimed to control for variations in brain volume that could be influenced by age or gender differences.

2.5. Procedures

The procedure began with participants undergoing the SNSB-C, followed by an assessment from two experienced neurologists who determined the presence of MCI based on Albert et al.'s guidelines, utilizing the SNSB-C results. Subsequently, participants engaged in the virtual kiosk test and underwent MRI imaging in an alternating order to eliminate any sequence bias. The same neurologists who diagnosed MCI also oversaw the administration of the virtual kiosk test, while a radiologist with 16 years of experience conducted the MRI scans. To ensure participants were comfortable with the VR environment, two introductory sessions were conducted prior to the virtual kiosk test. While the option for breaks was provided in case of discomfort, all participants were able to complete the study procedures without any need for interruption.

2.6. Analysis

Statistical evaluations in this study were executed utilizing IBM SPSS Statistics 27. We initiated the analysis by comparing demographic data between healthy controls and MCI patients, applying chi-square (χ^2) tests and t-tests for independent samples. Following this, analyses of covariance (ANCOVA) with age accounted for as a covariate were conducted to discern distinctions in neuropsychological traits, VR biomarkers, and MRI biomarkers between the two groups. Additionally, a Pearson correlation analysis was carried out to investigate the correlations among neuropsychological traits, VR biomarkers, and MRI biomarkers.

2.7. Machine learning

A machine learning model was crafted to classify subjects using neuropsychological traits, VR biomarkers, and MRI biomarkers constructed using Python 3. The model harnessed the capabilities of the Support Vector Machine (SVM) algorithm, particularly leveraging the Radial Basis Function kernel, with the parameters set to $C = 10$ and $\gamma = 1$. To ensure robust external validation, the dataset was divided into training and testing subsets, adhering to a 7:3 split. The efficacy of these models was gauged through a suite of performance metrics, including accuracy, sensitivity, specificity, precision, F1-score, and the Area Under the receiver operating characteristic Curve (AUC).

3. Results

3.1. Demographic characteristics

Chi-square tests revealed no significant variation in the gender distribution (male/female) between the two groups. Additionally, independent sample t-tests indicated that there were no statistically significant disparities in terms of age and education level among the participants.

Table 1: Demographics characteristics overview

	HC (N=22)	MCI (N=32)	<i>p</i> value
Male/Female	8/14	18/14	.246
Age	69.86±6.72	73.47±8.39	.100
Educational level	12.09±4.46	9.47±5.12	.057

3.2. Differences in Neuropsychological tests between Healthy Controls and MCI Patients

Compared to healthy controls, MCI patients exhibited substantial deficits across all five cognitive domains assessed: attention ($F_{1,52}=0.668$; $p<.001$), language function ($F_{1,52}=4.877$; $p<.001$), visuospatial function ($F_{1,52}=18.393$; $p<.001$), memory ($F_{1,52}=0.016$; $p<.001$), and frontal/executive function ($F_{1,52}=0.141$; $p<.001$).

Table 2: Summary of neuropsychological tests results

	HC (N=22)	MCI (N=32)	<i>p</i> value
DST: B	4.27±0.83	3.00±0.88	<.001
S-K-BNT	12.91±1.44	10.78±1.99	<.001
RCFT: CS	33.34±2.31	27.0±5.93	<.001
SVLT-E: DR	6.82±2.63	2.50±2.55	<.001
DSC	60.95±15.83	37.97±17.24	<.001

3.3. Differences in VR biomarkers between Healthy Controls and MCI Patients

Incorporating age as a covariate, the ANCOVA examination of VR biomarkers uncovered notable distinctions between healthy controls and MCI patients. Specifically, individuals with MCI demonstrated a markedly reduced speed of hand movement ($F_{1,52}=0.171$; $p<.001$), a shorter duration of fixation on target items ($F_{1,52}=1.641$; $p=.014$), and a higher incidence of errors ($F_{1,52}=10.529$; $p=.001$) in comparison to their healthy counterparts.

Table 3: Summary of VR biomarkers data

	HC (N=22)	MCI (N=32)	<i>p</i> value
Hand movement speed (m/s)	0.23±0.06	0.17±0.06	<.001
Proportion of fixation duration (%)	55.75±13.19	44.61±17.43	.014
The number of errors	1.74±1.61	4.00±2.81	.001

3.4. Differences in MRI biomarkers between Healthy Controls and MCI Patients

Utilizing ANCOVA with age as a control variable, we analyzed variations in ICV and its proportionate representation between healthy controls and MCI patients. Although MCI patients exhibited a larger ICV compared to healthy controls, the difference did not reach statistical significance. Notably, there was a significant reduction key brain regions in MCI patients, indicative of atrophy. The entorhinal cortex ($F_{1,52}=0.517$; $p=.001$), hippocampus ($F_{1,52}=2.188$; $p<.001$), and amygdala ($F_{1,52}=1.376$; $p=.002$) all showed significant differences, indicating notable atrophy in these regions.

Table 4: Summary of MRI biomarkers data

	HC (N=22)	MCI (N=32)	<i>p</i> value
ICV (cc)	1490.96 ±127.40	1511.49 ±128.89	.566
Amygdala (%)	23.57±1.99	21.42±2.44	.001
Hippocampus (%)	47.75±4.49	42.38±5.48	<.001
Entorhinal cortex (%)	27.71±2.56	24.81±3.68	.002

3.5. Correlation between Neuropsychological tests, VR biomarkers, and MRI biomarkers

A Pearson correlation analysis was performed to investigate the associations among neuropsychological traits, VR biomarkers, and MRI biomarkers. Each VR biomarker showed significant correlations with at least four, if not all, of the neuropsychological assessments. The correlation of hand movement speed with cognitive functions was especially noteworthy: attention ($r=.416$, $p=.002$), language ($r=.473$, $p<.001$), visuospatial ($r=.429$, $p=.001$), memory ($r=.313$, $p=.021$), and frontal/executive ($r=.652$, $p<.001$). Furthermore, each VR biomarker was found to have significant correlations with various MRI biomarkers. Of particular note was the correlation of fixation duration with all assessed MRI biomarkers: amygdala ($r=.275$, $p=.044$), hippocampus ($r=.335$, $p=.013$), and entorhinal cortex ($r=.281$, $p=.039$).

3.6. Machine learning performance

Analysis of SVM model performance showed that among individual modalities, neuropsychological traits led in accuracy at 90.7%, with VR biomarkers close behind at 87.0%, and MRI biomarkers at 83.3%. When modalities were paired, the combination of VR and MRI biomarkers outshone individual metrics, achieving an impressive 92.6% accuracy, along with 100.0% sensitivity, 92.6% specificity, 88.9% precision, a 94.1% F1 score, and an AUC of 0.91. The integration of VR biomarkers with neuropsychological tests, however, delivered the highest performance metrics, securing a 94.4% accuracy, 93.8% sensitivity, 94.4% specificity, 96.8% precision, a 95.2% F1 score, and an AUC of 0.93.

4. Discussion

The aim of this study was to bolster the clinical relevance of early MCI detection methods by incorporating VR biomarkers alongside traditional diagnostics. We observed that, in comparison to healthy controls, MCI patients displayed notably slower hand movements, shorter fixation times, and a higher error count when assessed with VR biomarkers. Noteworthy too was the discovery of strong associations between VR biomarkers and both MRI biomarkers and neuropsychological metrics. Intriguingly, the VR biomarkers alone achieved a high accuracy rate of 87% in detecting MCI, closely rivaling the 90.7% accuracy afforded by the neuropsychological measures, which served as the benchmark in this study. The combination of VR biomarkers and neuropsychological assessments further enhanced the diagnostic performance, reaching an accuracy of 94.4%, with sensitivity at 93.8%, specificity at 94.4%, precision at 96.8%, an F1 score of 95.2%, and an AUC of 0.93. These results underscore the augmented diagnostic efficacy

achieved by integrating VR biomarkers with either MRI biomarkers or neuropsychological assessments, as opposed to relying on the latter two in isolation.

5. Conclusion

This study's results highlight the effectiveness of VR biomarkers as a standalone tool for the early detection of MCI, showcasing their robust performance. Moreover, when integrated with conventional biomarkers, VR biomarkers offer the possibility to further elevate diagnostic accuracy. This suggests that the most efficacious strategy for early MCI diagnosis may involve the complementary use of VR with either MRI or neuropsychological assessments. Nonetheless, the findings also point to the potential of VR biomarkers as an independent diagnostic tool, offering a promising alternative to mitigate the financial burdens associated with MRI and the time constraints inherent in neuropsychological testing.

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